Ship Detection in SAR Images via Local Contrast of Fisher Vectors

Xueqian Wang, Gang Li, *Senior Member, IEEE*, Xiao-Ping Zhang, *Senior Member, IEEE*, and You He

***Abstract*— Existing superpixel-based detection algorithms for ship targets in synthetic aperture radar (SAR) images are often derived from the local contrast of intensities (i.e., the local contrast of the first-order information of superpixels) leading to deteriorating performance in low signal-to-clutter ratio (SCR) cases due to the low contrast between the intensities of targets and the clutter. In this article, we propose a new superpixel-based detector to improve the performance of ship target detection in SAR images via the local contrast of fisher vectors (LCFVs). The new LCFV-based detector exploits multiorder features of the superpixels based on the Gaussian mixture model (GMM) and accordingly improves the discrimination capability between the ship targets and the sea clutter, especially in low SCR cases. Experimental results demonstrate that the proposed LCFV-based detection algorithm provides better detection performance than the commonly used detection algorithms.**

***Index Terms*— Fisher vectors (FVs), local contrast, ship detection, superpixels, synthetic aperture radar (SAR).**

I. Introduction

AS AN active sensor for microwave imaging, synthetic aperture radar (SAR) is capable of providing the high-resolution image of targets. The imaging process of SAR is less affected by light and weather conditions than the passive infrared and optical imaging sensors. In other words, SAR can work in all-weather and all-day conditions. In the last decades, spaceborne and airborne SAR systems have gained extensive development and accompanied by the rising demand in military and civilian applications [1]–[5].

Recently, the problem of ship detection in SAR images has received much attention due to its important role in ocean monitoring/management [6]–[17], [29], [32]. Among the existing ship detection algorithms, constant false alarm rate (CFAR)-based algorithms have been widely investigated [7]–[13]. In [7], a two-parameter CFAR detector is developed and works well in single-target scenarios, where the sea clutter is characterized by its mean value and variance. In [8] and [9], CFAR detectors based on order-statistics are proposed to maintain good detection performance in multiple-target cases. Inspired by the pioneering work in [7]–[9], some variants of CFAR have been proposed to improve the detection performance from different aspects [10]–[13]. In [10], the difference between targets and sea clutter is analyzed based on the correlation of intensities in target regions and the log-normal distribution of the sea clutter. In [11], the Parzen-window kernel is utilized to estimate the intensity distribution of SAR images. Both the intensity distribution and the spatial distribution of SAR images are considered in [12]. Dai *et al*. [13] use the strategy of object proposal to extract the feature of ship targets with different sizes. Despite the superiority of adaptive decision thresholds as shown in [7]–[13], the above detection strategies based on CFAR mainly pay attention to the strong scattering targets [14], [57].

In addition to CFAR detectors mentioned above, some detectors are also proposed by further exploiting the pixel-level or rectangular-window-level characteristics of ship targets and the sea clutter [18]–[26]. In [18] and [19], the sparse feature of ship targets in SAR images is extracted and the problem of ship detection is converted to the problem of sparse signal recovery. In [20] and [21], standard deviation and kurtosis analysis of SAR images are presented to show the difference of ship targets and sea clutter, respectively. In [24], the characteristic of scattering tails of ship targets on the background is utilized to achieve the target detection. In [26], the entropy information of intensity in SAR images is used to enhance the contrast between the ship targets the sea clutter. Note that a ship target often occupies a connected region with an irregular shape in SAR images instead of an isolated pixel point or a region with a rectangular shape [14]. The above algorithms for ship detection [18]–[26] based on pixel-level or rectangular-window-level features are unable to exploit the boundary and structure information of ship targets. In particular, it is difficult for the detection algorithms based on pixel-level features [18]–[21] to distinguish whether a group of adjacent pixels to be tested belongs to the same ship target or the sea background [27].

Superpixels are visually meaningful and locally coherent regions, which reveal the boundary and structure information on the image [28]. In recent years, some superpixel segmentation algorithms have been developed for SAR images [35], [58]–[60]. In [35], the classical simple linear iterative clustering (SLIC) algorithm [28] is modified to make it suitable for intensity SAR images. In [58], the Gaussian

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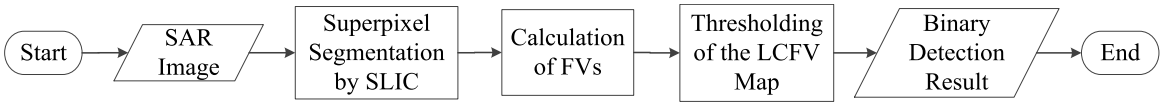


Fig. 1. Flowchart of the proposed LCFV-based detector.

kernel filtering is used to reduce the speckle noise in SAR images. Xiang *et al.* [59], [60] reduce the geometric distortions in SAR images and adaptively adjust the weights of features used for superpixel segmentation. Since superpixels can maintain the target boundary and reduce the speckle noise, some recent works of literature [14], [27], [29]–[31], [41] have been devoted to enhancing the detection performance of ship targets in SAR images based on the superpixel segmentation instead of the pixel-level analysis. The strategy of local contrast of intensity between the ship target superpixels and its surrounding sea clutter superpixels is commonly considered in these superpixel-based detectors. However, the contrast of simple intensity information in [14], [27], [29]–[31], and [41] may affect the performance of ship target detection in low signal-to-clutter ratio (SCR) cases due to the faint contrast of target-to-clutter in SAR images. Therefore, it is necessary to employ more effective features to improve the detection performance of superpixel-based algorithms, especially in low SCR cases.

The feature constructed by the Fisher vector (FV) has shown its superiority in the context of optical image processing [34]. FV of a superpixel is calculated based on the gradient of the log-likelihood with respect to the universal Gaussian mixture model (GMM) parameters [33]. FV represents the score function of superpixels on the multiple Gaussian components and contains the zero-order, first-order, and second-order information of the superpixel. Therefore, FV is expected to represent more characteristics of the superpixel than the features using simple intensity values. The first application of FV to ship detection was presented in [35], where the problem of ship detection is considered as a binary classification problem and a linear classifier is trained based on labeled data for ship detection and good detection performance is achieved. Note that "linear classifier in [35]" and "detector in [35]" are exchangeable hereinafter. However, the detector in [35] relies on the elaborate training data and provides the deteriorative performance when the SCR of training data is not consistent with that of the testing data. In addition, the contextual information in SAR images, i.e., the fact that ship targets are often surrounded by the sea clutter, has not been considered in [35].

1) Main Contribution: In this article, we propose a new superpixel-based ship detector via the local contrast of FVs (LCFVs). First, FVs are obtained based on the superpixel segmentation of SAR images. We show that the FVs of target superpixels are significantly different from that of their surrounding clutter superpixels. In addition, clutter superpixels around the ship target share the similar FVs in terms of sign information and amplitude, despite the fact that some clutter superpixels may contain sidelobes or scattering tails of the ship target. Inspired by these observations, the LCFV indicator measuring the difference of contextual FVs is proposed to find ship targets. Compared with detectors based on local contrast of intensities [14], [27], the proposed LCFV-based detector provides better detection performance, especially in low SCR cases. This benefits from the fact that more abundant differences (i.e., multi-(Gaussian) component and multiorder differences) between the ship targets and its surrounding sea clutter are exploited by using the contrast of contextual FVs. In addition, compared with the detector in [35], the proposed LCFV-based detector does not require the training data thanks to the exploiting of contextual information. Experimental results demonstrate that the proposed LCFV-based detection algorithm outperforms the state-of-the-art algorithms derived from local contrast [14], [27], especially in low SCR cases. In addition, the proposed LCFV-based detector without training data provides better detection performance than the linear classifier proposed in [35], when the SCR of training data for the classifier in [35] is mismatched with that of testing images. In other words, the proposed LCFV-based detector is more robust to the SCR of the testing SAR image compared with the detector in [35].

The remainder of this article is organized as follows. Section II formulates the proposed LCFV-based ship detector. In Section III, we state the superiority of the proposed detector compared with existing detectors. In Section IV, the effectiveness of the proposed detection algorithm is evaluated by using SAR images collected by Sentinel-1 and Gaofen-3 satellites. Finally, concluding remarks are given in Section V.

II. New Ship Detection Algorithm via Local Contrast of Fisher Vectors

Existing superpixel-based ship detectors derived from the strategy of local contrast mainly consider the difference of intensities between target and clutter [14], [27]. These detectors do not perform well in low SCR cases due to the small gap of intensities between the targets and the background clutter. In this section, we formulate the proposed ship detector based on LCFV to improve over the existing detection methods, especially in low SCR cases. The flowchart of the proposed ship detector is shown in Fig. 1. Four steps are contained in the proposed method. First, the superpixel segmentation of the SAR image is conducted to obtain locally coherent and compact regions. Then, FVs are calculated as the coding vectors of superpixels based on GMM of the SAR image. Next, LCFV is proposed to measure the multicomponent and multiorder contrast between the target superpixel and its neighboring clutter superpixels. The abundant contextual differences captured by LCFV effectively enhance weak ship targets and suppress the sea clutter. Finally, thresholding is operated on the LCFV map to generate binary detection results. In the following, we present the details of the above steps.

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Algorithm 1 SLIC for the SAR Image

Input: The testing SAR image, superpixel size S, regularization parameter λ.

Step-1: Initialize the center coordinate $[x_{(l)}, a_{(l)}, b_{(l)}]$ of l-th superpixel with the regular grid interval S, $\forall l$. Then, move these centers to the positions with lowest gradient in their 3×3 neighborhoods to avoid centering a superpixel on the boundary.

Step-2: The standard k-means algorithm is applied to associate each pixel with its nearest cluster center by using the distance measure in (1). The search area for each superpixel center is $2S \times 2S$.

Step-3: Each cluster center is updated by calculating the mean coordinate of all the pixels corresponding to it.

Step-4: Repeat Step-2 and Step-3 until the change of center coordinates is negligible.

Step-5: Postprocessing: "orphaned pixels" [28] are assigned the label of its nearest superpixel to enforce the connectivity.

Output: Superpixels with their corresponding local pixels.

A. Step-1: Superpixel Segmentation

The SLIC algorithm in [28] and [35] is used to achieve the superpixel segmentation due to its simplicity and the capability of maintaining the boundaries.

Assume the number of all the pixels in the SAR image is N and the size of superpixel is S. The expected number of superpixels in the image is $L = N/S^2$. First, the distance measure $D(i, j)$ between the pixel i and the pixel j in the SAR image is defined as [35]

$$D(i, j) = \sqrt{\left(\frac{\lambda}{S}\right)^2 \left[\left(a_i - a_j\right)^2 + \left(b_i - b_j\right)^2\right] + \left(x_i - x_j\right)^2} \tag{1}$$

where $i, j \in \{1, 2, \ldots, N\}$, (a_i, b_i) and $\left(a_j, b_j\right)$ are spatial coordinates of the pixel i and the pixel j, respectively, x_i and x_j denote the corresponding intensity values, λ is the regularization parameter that takes a balance between the spatial dissimilarity and the intensity dissimilarity. The SLIC algorithm [28] for the superpixel segmentation on SAR images is summarized in Algorithm 1. The lth superpixel obtained by the above SLIC algorithm is denoted by $\mathbf{x}_l = [x_{l,1}, x_{l,2}, \ldots, x_{l,P_l}]^T$, where P_l is the number of pixels in the lth superpixel, $(\cdot)^T$ is the transpose operation and $l = 1, 2, \ldots, L$. Note that the final number of superpixels in the image is slightly less than L sometimes, due to the postprocessing in Step-5 of SLIC. For convenience, we still use L to denote the final number of superpixels in the SAR image throughout this article. In Section IV, the parameter selection for SLIC in Algorithm 1 is discussed.

B. Step-2: Calculation of FVs

In this section, we calculate the FVs of superpixels. The basic idea of FV [33] is to first construct a generative GMM of the whole image, and then the gradients of the log-likelihood of superpixels with respect to all the model parameters are considered as the coding vectors of local features.

It is known that any continuous distributions can be approximated by GMM with arbitrary precision [54]. Here, a Q-component GMM with the parameter set $\mathbf{\Lambda} = \{\omega_q, \mu_q, \sigma_q, q = 1, 2, \ldots, Q\}$ is applied to model the SAR image, where ω_q, μ_q, and σ_q are the mixture weight, mean value, and standard deviation of the qth Gaussian component, respectively. The probability density function (PDF) of GMM can be written as

$$f_{\mathbf{\Lambda}}(x) = \sum_{q=1}^{Q} \omega_q \hat{f}_q(x) \tag{2}$$

where

$$\hat{f}_q(x) = \frac{1}{\sqrt{2\pi\sigma_q^2}} \exp\left[-\frac{1}{2\sigma_q^2}\left(x - \mu_q\right)^2\right]. \tag{3}$$

x represents the intensity value of the pixel and $\sum_{q=1}^{Q} \omega_q = 1$. The parameters in $\mathbf{\Lambda}$ for a testing SAR image can be estimated by using the expectation–maximization (EM) algorithm [47], which is based on the strategy of maximum likelihood estimation (MLE).

The score function $\mathbf{G}_{\mathbf{\Lambda}}(\mathbf{x}_l)$ of the lth superpixel $\mathbf{x}_l = [x_{l,1}, x_{l,2}, \ldots, x_{l,P_l}]^T$ with respect to the GMM parameter set $\mathbf{\Lambda}$ is

$$\mathbf{G}_{\mathbf{\Lambda}}(\mathbf{x}_l) = \sum_{p=1}^{P_l} \nabla_{\mathbf{\Lambda}} \log f_{\mathbf{\Lambda}}\left(x_{l,p}\right) \tag{4}$$

where $\nabla_{\mathbf{\Lambda}}$ is the gradient operator with respect to a parameter set $\mathbf{\Lambda}$, $\mathbf{G}_{\mathbf{\Lambda}}(\mathbf{x}_l) \in \mathrm{R}^{3Q \times 1}$ and $l = 1, 2, \ldots, L$. Based on (4), FV [33] of the lth superpixel is calculated as

$$\boldsymbol{\alpha}_l = \Phi \mathbf{G}_{\mathbf{\Lambda}}(\mathbf{x}_l) \quad \forall l \tag{5}$$

where Φ is the Cholesky decomposition of the matrix $\mathbf{F}_{\mathbf{\Lambda}}^{-1}$, $\mathbf{F}_{\mathbf{\Lambda}} = \mathrm{E}_{\mathbf{x}_l}\left\{\mathbf{G}_{\mathbf{\Lambda}}(\mathbf{x}_l)\left[\mathbf{G}_{\mathbf{\Lambda}}(\mathbf{x}_l)\right]^T\right\}$, and $\mathrm{E}_{\mathbf{x}_l}(\cdot)$ denotes the statistical expectation in terms of $\mathbf{x}_l$. The detailed formulation of the FV $\boldsymbol{\alpha}_l$ is given by

$$\boldsymbol{\alpha}_l = [\alpha_{l,\omega_1}, \alpha_{l,\omega_2}, \ldots, \alpha_{l,\omega_Q}, \alpha_{l,\mu_1}, \alpha_{l,\mu_2}, \ldots, \alpha_{l,\mu_Q}, \alpha_{l,\sigma_1}, \alpha_{l,\sigma_2}, \ldots, \alpha_{l,\sigma_Q}]^T \tag{6}$$

where

$$\alpha_{l,\omega_q} = \frac{1}{P_l\sqrt{\beta_q}} \sum_{p=1}^{P_l} \left(\phi_{l,p}^{(q)} - \beta_q\right) \tag{7}$$

$$\alpha_{l,\mu_q} = \frac{1}{P_l\sqrt{\beta_q}} \sum_{p=1}^{P_l} \phi_{l,p}^{(q)} \left(\frac{x_{l,p} - \mu_q}{\sigma_q}\right) \tag{8}$$

$$\alpha_{l,\sigma_q} = \frac{1}{P_l\sqrt{2\beta_q}} \sum_{p=1}^{P_l} \phi_{l,p}^{(q)} \left[\frac{\left(x_{l,p} - \mu_q\right)^2}{\sigma_q^2} - 1\right] \tag{9}$$

and $\beta_q = \exp\left(\omega_q\right) / \left(\sum_{\tilde{q}=1}^{Q} \exp\left(\omega_{\tilde{q}}\right)\right)$, $\phi_{l,p}^{(q)} = \left(\omega_q \hat{f}_q\left(x_{l,p}\right)\right) / \left(\sum_{\tilde{q}=1}^{Q} \omega_{\tilde{q}} \hat{f}_{\tilde{q}}\left(x_{l,p}\right)\right)$, for $l = 1, 2, \ldots, L$, $q = 1, 2, \ldots, Q$, $p = 1, 2, \ldots, P_l$. Note that (7)–(9) can be deemed the zero-order, the first-order, and the second-order information of the

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lth superpixel, respectively. Finally, the signed square-rooting is performed to avoid the too sparse FVs [50]

$$\boldsymbol{\alpha}_l \leftarrow \text{sign}(\boldsymbol{\alpha}_l)\,|\boldsymbol{\alpha}_l|^{1/2} \tag{10}$$

where $\text{sign}(\cdot)$ and $|\cdot|^{1/2}$ are both element-wise operators and $\text{sign}(\cdot)$ is 1 if the provided value is positive and -1 otherwise. The reason why (10) is required is that too sparse FVs makes it difficult to compare the corresponding superpixels [50]. After (10), all the FVs are self-l_2-normalized [61] for a fair comparison in Section II-C [see (11)].

C. Step-3: LCFVs

In this section, we propose a new indicator termed by LCFV to measure the contrast of contextual FVs and capture the *multicomponent* and *multiorder* information of the ship target and its surrounding clutter. The proposed LCFV is defined as

$$\theta_l = \text{med}\left\{\|\boldsymbol{\alpha}_l - \boldsymbol{\alpha}_{l^*}\|_2^2, \quad \text{for all } l^* \in \mathcal{N}(l)\right\} \tag{11}$$

where $\boldsymbol{\alpha}_l$ is the FV of the lth superpixel, "med" denotes the median value, $\mathcal{N}(l)$ denotes the index set of superpixels neighboring the lth superpixel, $l = 1, 2, \ldots, L$. In (11), the Euclidean distances $\{\|\boldsymbol{\alpha}_l - \boldsymbol{\alpha}_{l^*}\|_2^2, \ \forall\, l^*\}$ between FVs of the lth superpixel and its neighboring superpixels, i.e., for all $l^* \in \mathcal{N}(l)$, are first calculated. Then, the median value of Euclidean distances $\{\|\boldsymbol{\alpha}_l - \boldsymbol{\alpha}_{l^*}\|_2^2, \ \forall\, l^*\}$ is selected as the LCFV value of the current superpixel. The LCFV map of the SAR image is obtained by calculating the LCFV indicator for each superpixel in the image.

From (11), we can see that LCFV is based on the contextual differences of FVs. Existing works of literature involved with FV representation [33]–[35], [50], [61] have shown that different kinds of objects often have significantly dissimilar FVs, while objects belonging to the same class share the internally similar FVs. Therefore, when the lth superpixel to be tested is the target superpixel, the LCFV indicator θ_l in (11) tends to be a large value due to the dissimilarity between FVs of ship target superpixels and their surrounding sea clutter superpixels. In addition, when the lth superpixel belongs to the background clutter, θ_l is a small value because all the sea clutter superpixels share similar FVs. In Section III-A, we will further analyze the effectiveness of LCFV in (11) in terms of target enhancement, clutter suppressing, and robustness for the problem of ship target detection.

D. Step-4: Thresholding Operation

Thresholding operation can be performed on the LCFV map to obtain the binary detection results. A strategy for the selection of threshold is based on the PDF of the LCFV map, such as thresholding operations in CFAR detectors or the bright target segmentation [30], [56]. However, it is difficult to obtain the analytical PDF of LCFV in (11). In this article, the global threshold η is adaptively determined by using the classical two-parameter method [55], i.e.,

$$\eta = \bar{\theta}_{\text{LCFV}} + \xi\sigma_{\text{LCFV}} \tag{12}$$

where

$$\bar{\theta}_{\text{LCFV}} = \frac{1}{L}\sum_{l=1}^{L}\theta_l$$

$$\sigma_{\text{LCFV}} = \sqrt{\frac{1}{L}\sum_{l=1}^{L}\left(\theta_l - \bar{\theta}_{\text{LCFV}}\right)^2}$$

are the mean value and the standard deviation of $\{\theta_l, l = 1, 2, \ldots, L\}$, respectively, ξ is the adjustment factor. In practice, the parameter ξ often ranges from 3 to 14 [51]. The binary detection results $\{\delta_l, l = 1, 2, \ldots, L\}$ can be generated by

$$\begin{cases}\delta_l = 1, & \text{if } \theta_l > \eta \\ \delta_l = 0, & \text{if } \theta_l \le \eta\end{cases} \tag{13}$$

for $l = 1, 2, \ldots, L$. In (13), $\delta_l = 1$ represents that ship target is presented at the lth superpixel, while sea clutter is presented with $\delta_l = 0$, for $l = 1, 2, \ldots, L$.

The implementation of the proposed LCFV-based detector is summarized in Algorithm 2.

Algorithm 2 The Proposed LCFV-Based Detector

Input: The testing SAR image, superpixel size S, regularization parameter λ, the number of Gaussian components Q for GMM.

Step-1: Superpixel segmentation of the SAR image by using SLIC in Algorithm 1 with the superpixel size S and the regularization parameter λ. The l-th superpixel obtained by the SLIC algorithm is denoted by $\mathbf{x}_l$, $\forall l$.

Step-2: Estimate the GMM parameter set $\boldsymbol{\Lambda} = \{\omega_q, \mu_q, \sigma_q, q = 1, 2, \ldots, Q\}$ of the whole SAR image by using the EM algorithm. Then, according to (6)-(10) and the GMM parameter set $\boldsymbol{\Lambda}$, calculate FV $\boldsymbol{\alpha}_l$ for each superpixel $\mathbf{x}_l$ in the SAR image, for $l = 1, 2, \ldots, L$. Next, all the FVs are self-l_2-normalized.

Step-3: Calculation of the LCFV indicator $\theta_l = \text{med}\left\{\|\boldsymbol{\alpha}_l - \boldsymbol{\alpha}_{l^*}\|_2^2, \ \text{for all } l^* \in \mathcal{N}(l)\right\}$ in (11) based on the local difference between FVs of targets and their surrounding clutter, for $l = 1, 2, \ldots, L$.

Step-4: Thresholding is performed on the LCFV map by using (12) and (13). Then, obtain the binary detection results $\{\delta_l, l = 1, 2, \ldots, L\}$.

Output: The binary detection results $\{\delta_l, l = 1, 2, \ldots, L\}$.

III. Performance Analysis of LCFV and Comparison to Other Related Detectors

A. Performance Analysis of LCFV

In this section, the performance analysis of LCFV in (11) is provided with respect to target enhancement and clutter suppressing. We also show the robustness of LCFV to the abnormal elements in contextual FVs and the results of superpixel segmentation.

Target Enhancement and Clutter Suppressing: In Figs. 2–4, we first visualize the contextual difference

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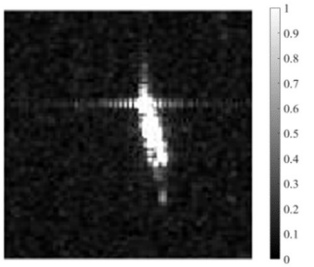


Fig. 2. Sentinel-1 intensity image of the Shanghai port area acquired on VH polarization channel [48]. The bright part in the center of the figure is a ship target.

of the FVs between the ship targets and its surrounding sea clutter. Fig. 2 shows an image slice collected by the Sentinel-1 satellite [48]. In Fig. 3(a), sea clutter level followed by the K-distribution [49] is added on the SAR image in Fig. 2 to conduct the anticlutter analysis, where SCR = 15 dB. It is worth noting that the proposed detector in this article does not rely on the K-distribution of the clutter. The simulated clutter in this article is only for the convenience of anticlutter analysis as in [23] or [35]. Other methods in [23] to generate the simulated clutter can also be used here. The definition of SCR and the choice of parameters in K-distribution are the same as those in the previous article in [35, Sec. IV. A]. Here, we consider the additive sea clutter as in [23] and [35]. The extensively investigated multiplicative clutter/noise in SAR images also can be converted as the additive clutter/noise as reported in [64]. In Fig. 3(b), the superpixel segmentation result of Fig. 3(a) is provided, where the target outline is well maintained. FVs of target superpixels and its five surrounding clutter superpixels {①, ②, ③, ④, ⑤} in Fig. 3(b) are used to show contextual differences among FVs. In Fig. 3(c), the zero-order, first-order, and second-order information in FVs, i.e., the subvectors of FVs with respect to the weight, mean value, and standard deviation weigh in GMM are illustrated at the first, second, and third columns, respectively. The red bars in the first row denote the elements in target FVs. The blue bars in the second, third, fourth, fifth, and sixth rows represent the elements in FVs of five surrounding clutter superpixels {①, ②, ③, ④, ⑤}, respectively. The horizontal axis $[1, 2, \ldots, 7]$ of each subfigure in Fig. 3(c) denotes the index of Gaussian components in GMM. From most of the values of FVs in Fig. 3(c) (e.g., first, third–seventh components at the first column, all the components at the second column, first, third–fifth, and seventh components at the third column), we can observe the following.

i) Elements in the target FV are significantly different from those in surrounding clutter FVs. The difference can be categorized into twofold: contrary signs or zero/negligible values in the target FV while dominant values in clutter FVs.

ii) Clutter superpixels share similar FVs in terms of the sign information and amplitude.

In Fig. 4, we further investigate the contrast of contextual FVs, where the target-to-clutter contrast is much lower than

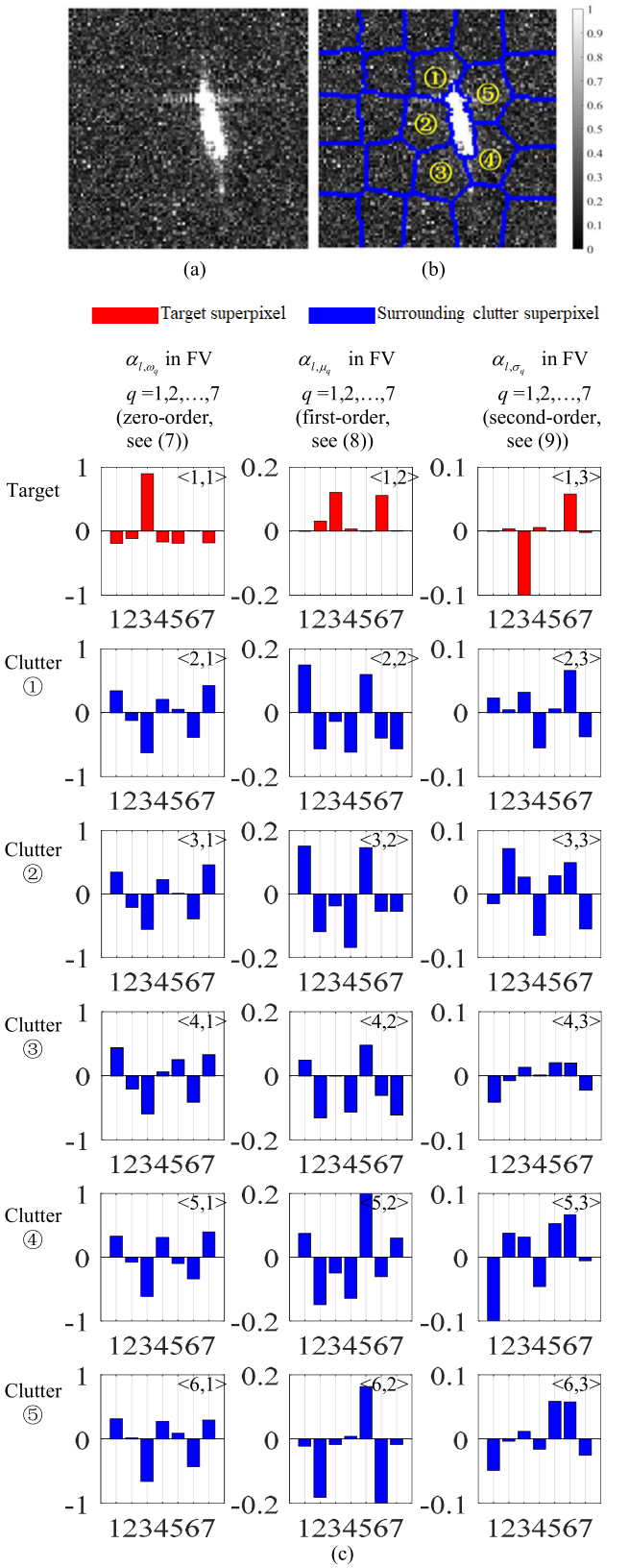


Fig. 3. (a) Intensity SAR image in Fig. 2 corrupted by the clutter (SCR = 15 dB). (b) Result of superpixel segmentation. The target superpixel and its five surrounding clutter superpixels {①, ②, ③, ④, ⑤} in (b) are illustrated for the following visualization of contextual differences among FVs. (c) Local contrast of FVs. The number of Gaussian components in GMM is $Q = 7$.

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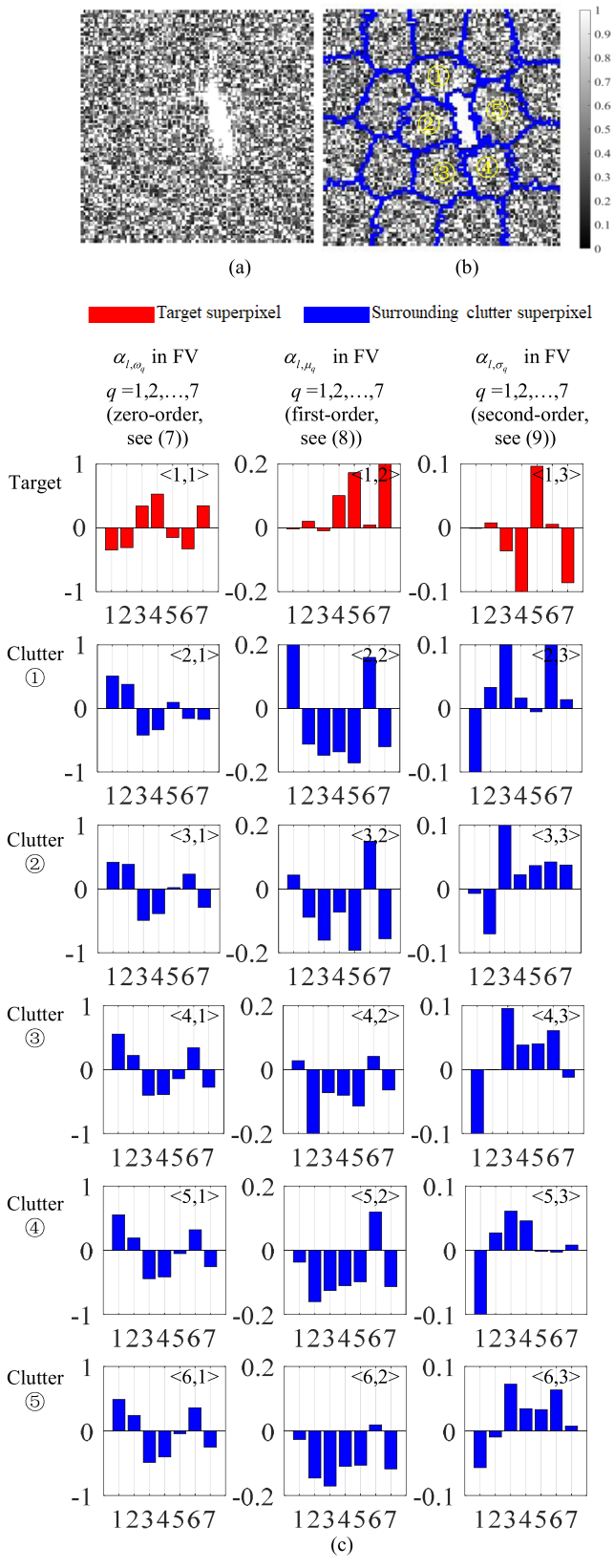


Fig. 4. (a) Intensity SAR image in Fig. 2 corrupted by the clutter (SCR = 3 dB). (b) Result of superpixel segmentation. The target superpixel and its five surrounding clutter superpixels {①, ②, ③, ④, ⑤} in (b) are illustrated for the following visualization of contextual differences among FVs. (c) Local contrast of FVs. The number of Gaussian components in GMM is $Q = 7$.

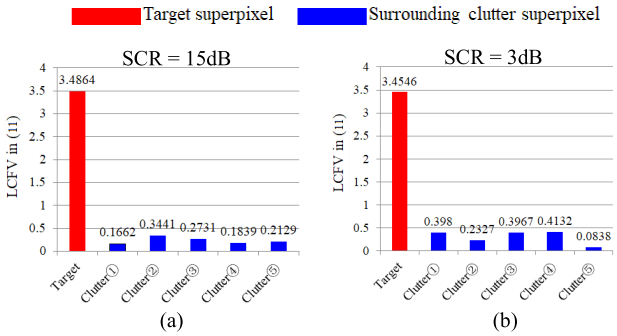


Fig. 5. LCFV values of the target superpixel and its five surrounding superpixels in (a) Fig. 3, SCR = 15 dB and (b) Fig. 4, SCR = 3 dB. LCFV is calculated after the square-rooting and self-l_2-normalization of FVs.

that in Fig. 3. In Fig. 4(c), the dissimilarity of FVs between the target and its surrounding clutter and the internal similarity of FVs among the clutter superpixels can also be observed. The definition of LCFV in (11) is mainly derived from the above features of contextual FVs presented in i) and ii).

On the one hand, LCFV considers the differences among all the elements in the contextual FVs. This is helpful to accumulate differences of multiorder information between targets and their surrounding clutter on most Gaussian components [see feature i)]. The accumulation of abundant differences leads to a large value of the LCFV indicator θ_l, i.e., enhanced ship target.

On the other hand, clutter superpixels around the ship target share the similarity of the multiorder information on most Gaussian components [see feature ii)]. This similarity results in a small value of the LCFV indicator θ_l, i.e., sea clutter can be effectively suppressed using the thresholding operation.

In Fig. 5, we show the LCFV values [calculated by (11)] of the target superpixel and its five surrounding clutter superpixels which are illustrated in Figs. 3 and 4. It is observed that the LCFV value of the target superpixel is dramatically higher than that of background clutter superpixels even in the low SCR case. In other words, weak targets can be effectively enhanced with suppressed clutter by using LCFV.

Robustness to the Abnormal Elements in Contextual FVs: Note that only a few elements in contextual FVs do not satisfy the feature i) or ii). For example, the second component of the target and clutter FVs in the first column of Fig. 3(c) have close values, which may be caused by the small gap of intensities between target and strong clutter points. The second component of clutter FVs in the third column of Fig. 3(c) do not have consistent sign information. This is because some clutter superpixels may contain the strong scattering tails or the sidelobe of targets, while other clutter superpixels are pure sea clutter. In (11), LCFV is robust to these "abnormal" elements because: 1) the proportion of these "abnormal" elements in FVs is small; 2) the amplitudes of these "abnormal" elements are relatively much smaller than that of "normal" elements [see the comparison of the third components at the first column and the second components at the third column of Fig. 3(c)];

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and 3) LCFV compares all the elements in contextual FVs and aggregates all the differences.

Median Filtering:*Robustness to Superpixel Segmentation:* The selection of the median value in (11) is robust to the results of superpixel segmentation. In other words, the median value of $\left\{\|\boldsymbol{\alpha}_l - \boldsymbol{\alpha}_{l^*}\|_2^2, \text{for all } l^* \in N(l)\right\}$ in (11) is still large, when the testing superpixel is part of a large ship target. In addition, the selection of median value in (11) prevents the clutter superpixels around the target superpixel from being false alarms. For example, when clutter superpixel "②" in Fig. 3(b) is the testing superpixel, the LCFV indicator of "②" is a small value thanks to the median filtering, even if there is a target superpixel around it.

B. Comparison With the Related Detectors

Wang *et al.* [14] and Yu *et al.* [27] consider the local contrast of intensities in the superpixels, i.e., only the first-order information is utilized. In addition, the simple contrast of intensities in [14] and [27] cannot provide "layered" information like multicomponent representation in this article. In low SCR cases, the intensities of ship target superpixels are close to that of their surrounding clutter superpixels. The proposed LCFV in (11) extracts more abundant characteristics (i.e., multiorder and multicomponent characteristics) of ship targets and sea clutter than the simple contrast of intensities used in [14] and [27]. Therefore, the proposed LCFV-based detector is expected to have better discrimination ability between targets and clutter than detectors in [14] and [27], especially for weak ship target detection.

Note that the proposed detector is different from that in [35], where a linear classifier is trained based on the labeled target/clutter superpixel samples and their FV representation. It is assumed in [35] that GMM of the training data is the same as that of the testing data. However, the change of SCR of SAR images leads to different parameters of GMM. Therefore, the detector in [35] may suffer from performance degradation, when the SCR of testing data is different from that of training data. In addition, the contextual information in the SAR images has not been considered in [35]. Compared with the detector in [35], the proposed LCFV-based detector exploits the contextual information in the SAR image and does not require the training data. Accordingly, the LCFV-based detector is more robust to the SCR of the testing SAR image than the detector in [35].

We also note that deep learning techniques are recently applied to the detection of ship targets in SAR images [36]–[40], [52], [57]. Superior detection performance and automatic feature extraction are often achieved by means of deep learning [57]. These ship detectors in [36]–[40], [52], and [57] require lots of labeled data. Accurate labeling is a heavy workload and the training process is time-consuming. In addition, when labeled data is not sufficient and diverse, they cannot provide informative guidance for the testing step. In this article, the proposed LCFV-based algorithm only considers the detection problem without training data as in [14], [18]–[21], and [27].

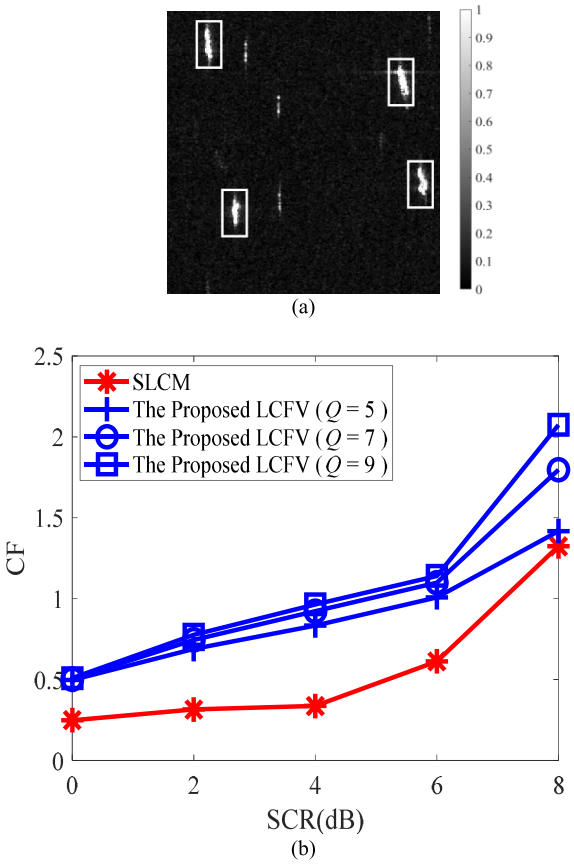


Fig. 6. (a) Sentinel-1 intensity image of Shanghai port area [48]. The ground truth of ship targets is marked by white rectangle boxes. (b) CF versus SCR generated by SLCM [14] and the proposed LCFV by using the SAR image in (a).

Another popular strategy to enhance the detection performance of ship or other marine targets is to increase the number of polarimetric channels, i.e., using the polarimetric SAR (PolSAR) [43]–[46]. Despite the better performance with PolSAR images, it is also important to improve the detection performance by utilizing single-polarization SAR images. In this article, we focus on the problem of ship detection using single-polarization SAR images. In addition, the proposed ship detector in this article does not rely on any specific polarization requirement.

IV. Experimental Results

In this section, we compare the performances of SLCM [14], SP-CFAR [27], linear classifier [35], and the proposed LCFV-based detector for ship detection in SAR images.

Two Sentinel-1 SAR images and a Gaofen-3 SAR image [Figs. 6–8(a)] are used in experiments to show the superiority of the proposed method. The key information of these three SAR images is provided in Table I. The speed of wind on the sea surface is also an important parameter for marine SAR images. In Table I, the exact values of wind speed shown in Figs. 6(a), 7(a), and 8(a) are not provided because the effect of wind speed has been reflected by the SCR values. We refer readers to [62] and [63] for more details about the

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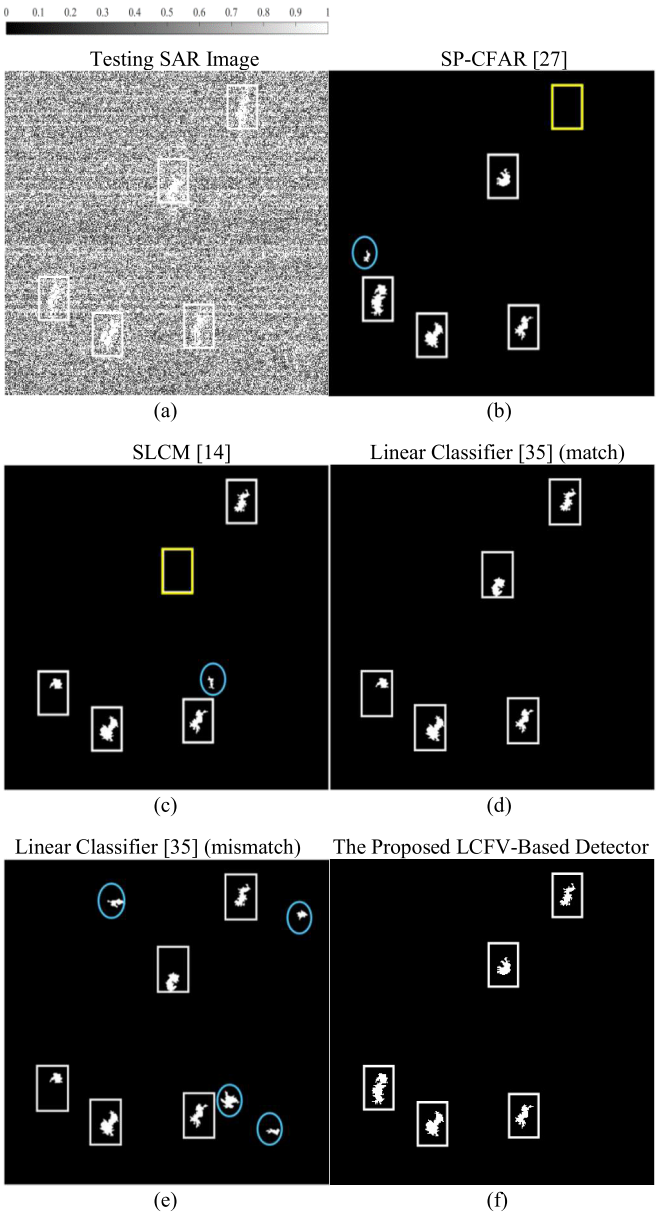


Fig. 7. (a) SAR intensity image from Gaofen-3, SCR = 5 dB. The ground truth of ship targets is marked by white rectangle boxes. The binary detection results of (b) SP-CFAR [27], (c) SLCM [14], (d) linear classifier [35], where the SCR of training data is matched with that of testing data, (e) linear classifier [35], where the SCR of training data (SCR = 3 dB) is mismatched with that of testing data, and (f) proposed LCFV-based detector. In (b)–(f), yellow boxes, white boxes, and blue ellipses denote the missed targets, correctly detected targets and false alarms, respectively.

effect of wind speed on the SAR images. The Sentinel-1 image in Fig. 6(a) contains four ship targets and has 310×300 pixels. The Sentinel-1 image in Fig. 8(a) has 2928×2693 pixels containing 92 ship targets. Multilook (five looks) preprocessing is used for Sentinel-1 images in Figs. 6(a) and 8(a). Radiometric calibration is also performed on these Sentinel-1 images by using the software SNAP 3.0 [66]. We refer the readers to [48] for complete details of the Sentinel-1 images in Figs. 6(a) and 8(a). The Gaofen-3 image in Fig. 7(a) with 381×361 pixels contains five ship targets. Preprocessing steps like multilook and radiometric calibration are not performed on the Gaofen-3 image in Fig. 7(a). All the SAR images in this article are obtained over the Shanghai port area. The ground truths of SAR images are obtained by combining the external automatic identification system (AIS) data and annotations from technical experts [48].

TABLE I
KEY INFORMATION RELATED TO THE SAR IMAGES USED FOR EXPERIMENTS IN SECTION IV

	Fig. 6(a) and Fig. 8(a)	Fig. 7(a)
Satellite	Sentinel-1	Gaofen-3
Polarization	VH	HH
Frequency Band	C	C
Centre Frequency	5.405 GHz	5.4 GHz
Angle of Incidence	30°-46°	20°-50°
Resolution	10m×10m	4m×4m
Kind of product	Ground range detected (GRD)	Single look complex (SLC)

A. Parameter Selection for Superpixel Segmentation and the Related Detectors

The superpixel size S can be determined according to a fixed percentage (e.g., 25%) of the average number of pixels occupied by ship targets [14]. A large value of λ means that the superpixel tends to be regular, while a small value of λ emphasizes the local coherence of intensity values. In practice, a small value of λ is suggested (e.g., $\lambda \approx 1$) as demonstrated in [35]. Existing literature shows that a few iterations (e.g., 5–20 iterations) are sufficient for the SLIC algorithm to provide satisfactory segmentation results [27], [28]. According to the above guidance, we choose the size of superpixel $S = 18$ and the regularization parameter $\lambda = 0.8$ in all experiments. The maximum iteration count of SLIC is 20. The decision threshold in SP-CFAR [27] is calculated with the false alarm ratio 5×10^{-4}. The hyperparameters of SLCM are selected according to [14]. For the proposed LCFV-based detector, the adjustment factor ξ in the threshold [see (12)] is 7. For completeness, the performance of the proposed detector is also investigated with thresholds scaled by different adjustment factors (see Fig. 9).

B. Anticlutter Performance: Contrast Factor

First, we show the anticlutter performance of LCFV in (11) in terms of contrast factor (CF) compared with the local contrast measure used in SLCM, which only considers the contrast of intensities of superpixels (see [14, eq. (9)]). CF is the ratio of the "weakest" target superpixel to the "strongest" clutter superpixel before the final thresholding process in detectors. CF is calculated as

$$\mathrm{CF} \triangleq \frac{\min_w \{\zeta_w,\ w = 1, 2, \ldots, N_{\mathrm{Tar}}\}}{\max_l \{\varphi_l, l \in \mathrm{Clu}\}} \tag{14}$$

where φ_l represents the indicator in (11) for the proposed LCFV detector and the output of [14, (9)] for the SLCM detector in calculating corresponding CFs, respectively, N_{Tar} is the total number of real ship targets, $\zeta_w = \max_l \{\varphi_l, l \in \mathrm{Tar}_{(w)}\}$, "$\mathrm{Tar}_{(w)}$" and "Clu" denote the ground truth of wth ship target

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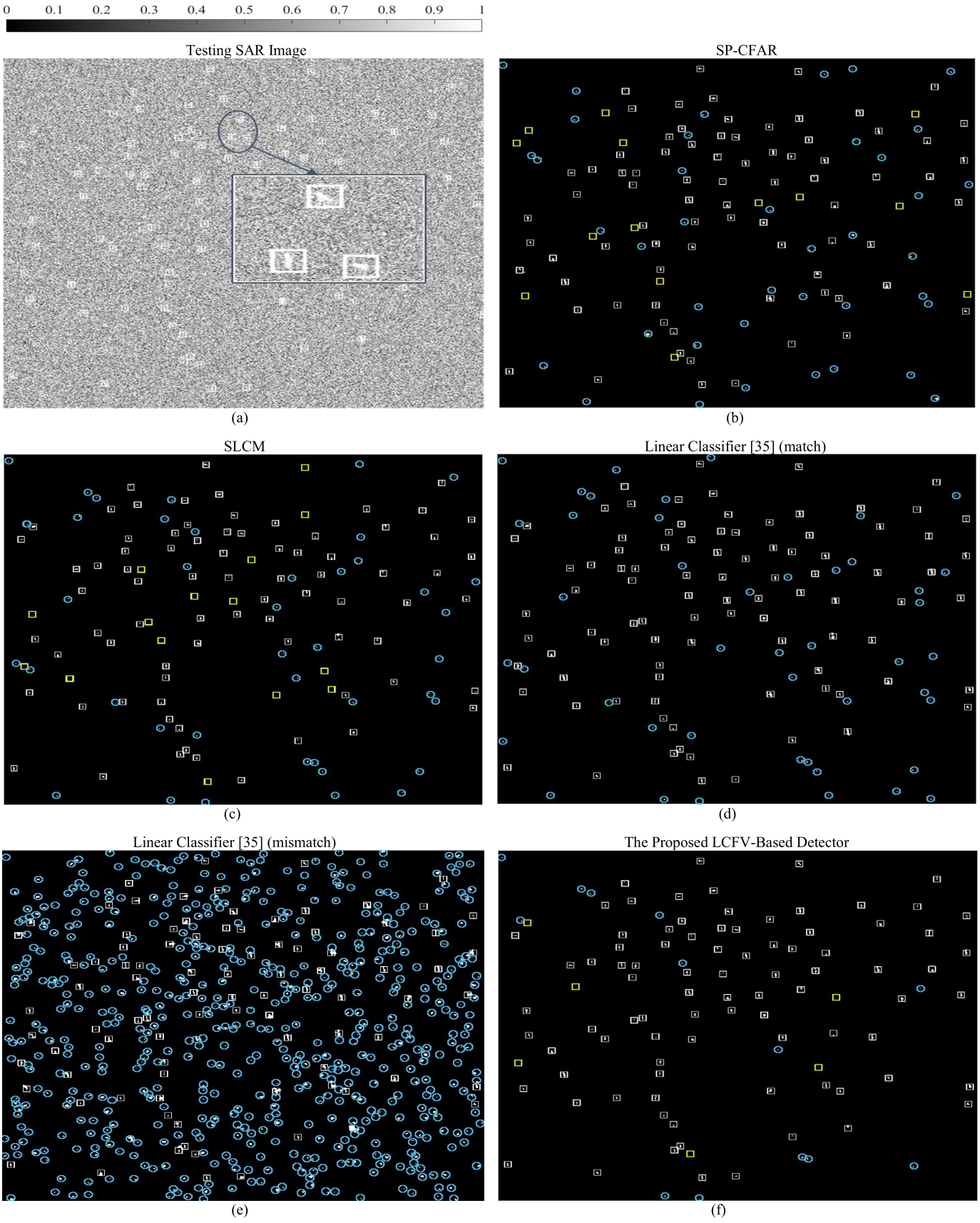


Fig. 8. (a) SAR intensity image from Sentinel-1, SCR = 0 dB. The ground truth of ship targets is marked by white rectangle boxes. The binary detection results of (b) SP-CFAR [27], (c) SLCM [14], (d) linear classifier [35], where the SCR of training data is matched with that of test data, (e) linear classifier [35], where the SCR of training data (SCR = 3 dB) is mismatched with that of test data, and (f) proposed LCFV-based detector. In (b)–(f), white boxes, yellow boxes, and blue ellipses represent the correctly detected targets, missed targets, and false alarms, respectively.

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and clutter areas in the SAR images, respectively. A large value of CF means enhanced targets and suppressed sea clutter, i.e., it is easy to further detect weak ship targets from the background clutter.

Here, five levels of clutter modeled by K-distribution (SCR is increased from 0 to 8 dB with a step size of 2 dB) are added on the Sentinel-1 image in Fig. 6(a) to generate simulated SAR images and show the superior performance of the proposed method in low SCR cases. As SCR increases, Fig. 6(b) shows the trends of CFs of SLCM and the proposed LCFV by using the SAR image in Fig. 6(a). CFs in Fig. 6(b) are obtained by averaging over 10^2 random trails. As illustrated in Fig. 6(b), the performance of LCFV is consistently better than that of SLCM in low SCR cases thanks to more abundant differences (i.e., multiorder and multicomponent differences) between targets and the clutter considered in LCFV. In Fig. 6(b), CFs of the proposed LCFV are also presented with different values of Q, i.e., the number of Gaussian components in GMM. From Fig. 6(b), we can see that LCFV shows better CF performance with a larger value of Q. Note that a large value of Q increases the computation burden of parameter estimation for GMM, while a too small value of Q may not effectively model the SAR image. Without loss of generality, $Q = 7$ is selected in the experiments hereinafter for the compromise between performance and complexity. We refer readers to [53] for more alternative methods to choose the value of Q.

C. Detection Performance

In this section, we compare the probability of detection (P_d), probability of false alarm (P_{fa}), and figure of merit (FoM) of SP-CFAR [27], SLCM [14], linear classifier [35] and the proposed LCFV-based detector by using two SAR images from Sentinel-1 and Gaofen-3 satellites, as shown in Figs. 7(a) and 8(a), respectively. All the ship targets shown in Figs. 7(a) and 8(a) are relatively weak with low SCRs. The performance metrics P_d, P_{fa}, and FoM are defined as

$$P_d = \frac{N_{\text{DecTar}}}{N_{\text{Tar}}} \tag{15}$$

$$P_{\text{fa}} = \frac{N_{\text{DecFal}}}{L - N_{\text{Tar}}} \tag{16}$$

$$\text{FoM} = \frac{N_{\text{DecTar}}}{N_{\text{DecFal}} + N_{\text{Tar}}} \tag{17}$$

respectively, where N_{DecTar} is the number of correctly detected ship targets, N_{DecFal} is the number of false alarms.

In Fig. 7(b)–(f), the missed targets, correctly detected targets, and false alarms are represented by the yellow boxes, white boxes, and blue ellipses, respectively. SP-CFAR [27] and SLCM [14] only consider the contrast of intensities between target and clutter superpixels. From Fig. 7(b) and (c), it is evident that SP-CFAR and SLCM suffer from some false alarms and undetected targets. In Fig. 7(d) and (e), binary detection results of a linear classifier in [35] are provided, where training samples containing two target superpixels and six clutter superpixels are used. The SCR of training samples for Fig. 7(d) is consistent with that of the testing data, while not for Fig. 7(e). We can see that the linear classifier in [35] can maintain good performance when SCRs of testing data and training data are consistent, however suffers from four false alarms with the mismatched SCR. In Fig. 7(f), the proposed LCFV-based detector outperforms the local-contrast-based detectors [14], [27] in the following two aspects: 1) more weak ship targets are correctly detected and 2) the sea clutter areas are significantly suppressed. Compared with the linear classifier in [35], the proposed LCFV-based detector is more robust to the SCR of the testing SAR image. Similarly, it is observed from Fig. 8(b)–(f) that the proposed LCFV-based detector provides a relatively clear background and finds more weak ship targets compared with SLCM [14] and SP-CFAR [27]. In addition, the proposed LCFV-based detector shows the better performance than the linear classifier [35] with the mismatched SCRs between the testing data and the training data (for Fig. 8(d) and (e), training samples with ten target superpixels and ten clutter superpixels are used).

TABLE II
P_d, P_{fa} AND FOM OF DIFFERENT DETECTORS

Scenarios / Detectors	Scene in Fig. 7 (Gaofen-3)			Scene in Fig. 8 (Sentinel-1)		
	P_{d}	P_{fa}	FoM	P_{d}	P_{fa}	FoM
SP-CFAR [27]	0.8000	0.0029	0.6667	0.8495	0.0023	0.5909
SLCM [14]	0.8000	0.0029	0.6667	0.8387	0.0023	0.5790
Linear Classifier [35] (match)	1.0000	0	1.0000	**1.0000**	0.0025	0.6788
Linear Classifier [35] (mismatch)	1.0000	0.0116	0.5555	1.0000	0.0323	0.1426
LCFV-Based Detector	**1.0000**	**0**	**1.0000**	0.9247	**0.0007**	**0.8095**

Quantitatively, compared with SP-CFAR [27] and SLCM [14], higher P_d, higher FoM, and lower P_{fa} are provided by the proposed LCFV-based detector, as shown in Table II. This indicates that the LCFV-based detector achieves better detection performance in low SCR cases thanks to the more effective contrast between contextual FVs of targets and sea clutter. In addition, compared with the linear classifier [35] in mismatched SCR scenarios, the proposed LCFV-based detector significantly improves the detection performance with higher FoM.

SLCM [14], SP-CFAR [27], and the proposed LCFV-based detector are based on the thresholding operation. To thoroughly investigate the detection performance, we plot the receiver operating characteristic (ROC) [65] curves in Fig. 9, where P_d and P_{fa} in ROC curves are calculated based on different thresholds. We can see that the proposed LCFV-based detector provides better global detection performance than SLCM [14] and SP-CFAR [27] on Gaofen-3 and Sentinel-1 images. Again, the superiority of the proposed LCFV-based detector is confirmed.

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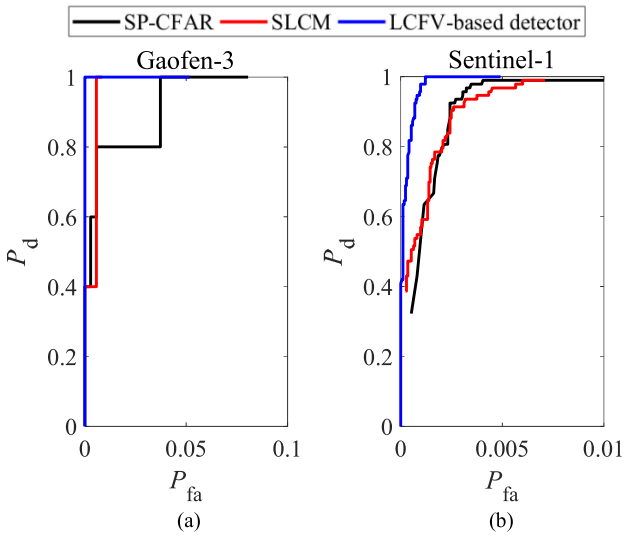


Fig. 9. ROC curves of SLCM [14], SP-CFAR [27], and the proposed LCFV-based detector. (a) Gaofen-3 image in Fig. 7. (b) Sentinel-1 image in Fig. 8.

TABLE III
COMPARISON OF THE RUNNING TIME (s)

Detectors / Running Times / Scenarios	SP-CFAR [27]	SLCM [14]	The Proposed LCFV-Based Detector
Scene in Fig. 7 (Gaofen-3)	0.0344	0.0054	0.0226
Scene in Fig. 8 (Sentinel-1)	3.0661	0.6520	0.7672

It is worth mentioning that extremely low SCR may cause the nonideal superpixel segmentation, i.e., corrosion of the ship outlines (see Fig. 7). Despite this, the detected target superpixels still contain most of the target pixels or strong backscattering parts of ship targets [14].

D. Running Time

In this section, we compare the running times of SLCM [14], SP-CFAR [27], and the proposed LCFV-based detector in Table III. All these detectors do not require the training process. For a fair comparison, the detector in [35] requiring training time is not considered here. Experiments are conducted in the 64-bit Windows system with the Intel Core i5 CPU and 16-GB RAM. The software platform is MATLAB 2019a. Running times are averaged over the total number of superpixels in the image. From Table III, we find that SP-CFAR has the maximum running time due to its time-consuming sliding window operation [27], [35]. The computation burden of SLCM [14] is lightest since it only calculates the mean values of superpixels. The computational time of the proposed LCFV-based detector is more than that of SLCM but less than that of SP-CFAR.

V. CONCLUSION

In this article, we consider the problem of ship detection in SAR images with superpixels. First, we proposed an LCFV indicator according to: 1) the dissimilarity of FVs between the target and its surrounding clutter and 2) the internal similarity of FVs among the clutter superpixels. LCFV dramatically enhances weak ship targets and suppresses the background clutter and is robust to the abnormal elements in contextual FVs and the results of superpixel segmentation. Based on the LCFV indicator, we proposed a new detection algorithm for ship detection. Unlike traditional detectors based on the local contrast of intensities of superpixels, the proposed LCFV-based detector considers multiorder and multicomponent contrast between targets and their surrounding clutter superpixels. In other words, the proposed detector exploits more abundant differences between ship target and background clutter. Therefore, the proposed detector can better distinguish target and clutter, especially in low SCR cases. In addition, compared with other FV-based detectors depending on the elaborate training data, the proposed new detector does not require the training process and is insensitive to the SCR of testing data thanks to the exploiting of contextual information.

Experimental results demonstrate that, compared with existing local-contrast-based detectors, the proposed detector offers enhanced detection performance with a higher probability of detection, higher FoM, and lower probability of false alarm in low SCR cases. Compared with other FV-based detectors, the proposed LCFV-based detector is more robust to the SCR of the testing SAR image. The combination of LCFV with the inshore ship detection in the more complex background will be pursued in our future research.

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